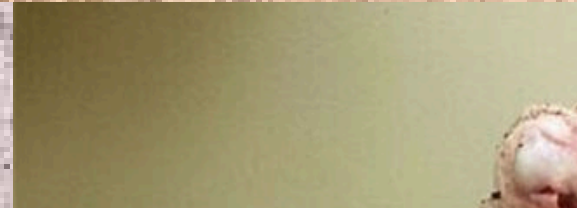
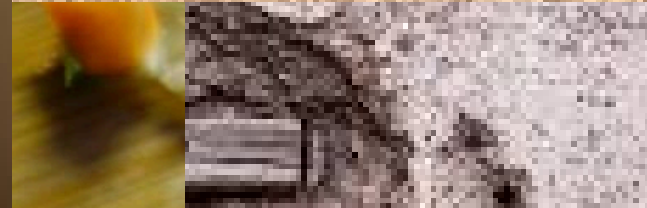




PAREIDOLIA

Is that a face?





DETECCIÓN DE PAREIDOLIA

TRANSFER LEARNING



PLANTEAMIENTO

1

Modelos
preentrenados

2

Transferir
conocimiento

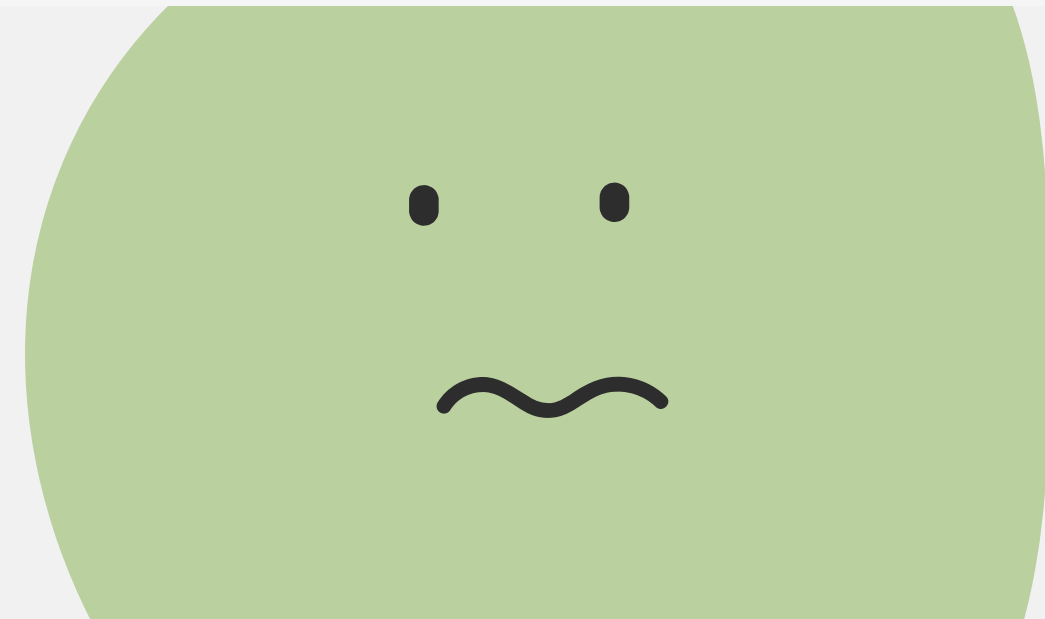
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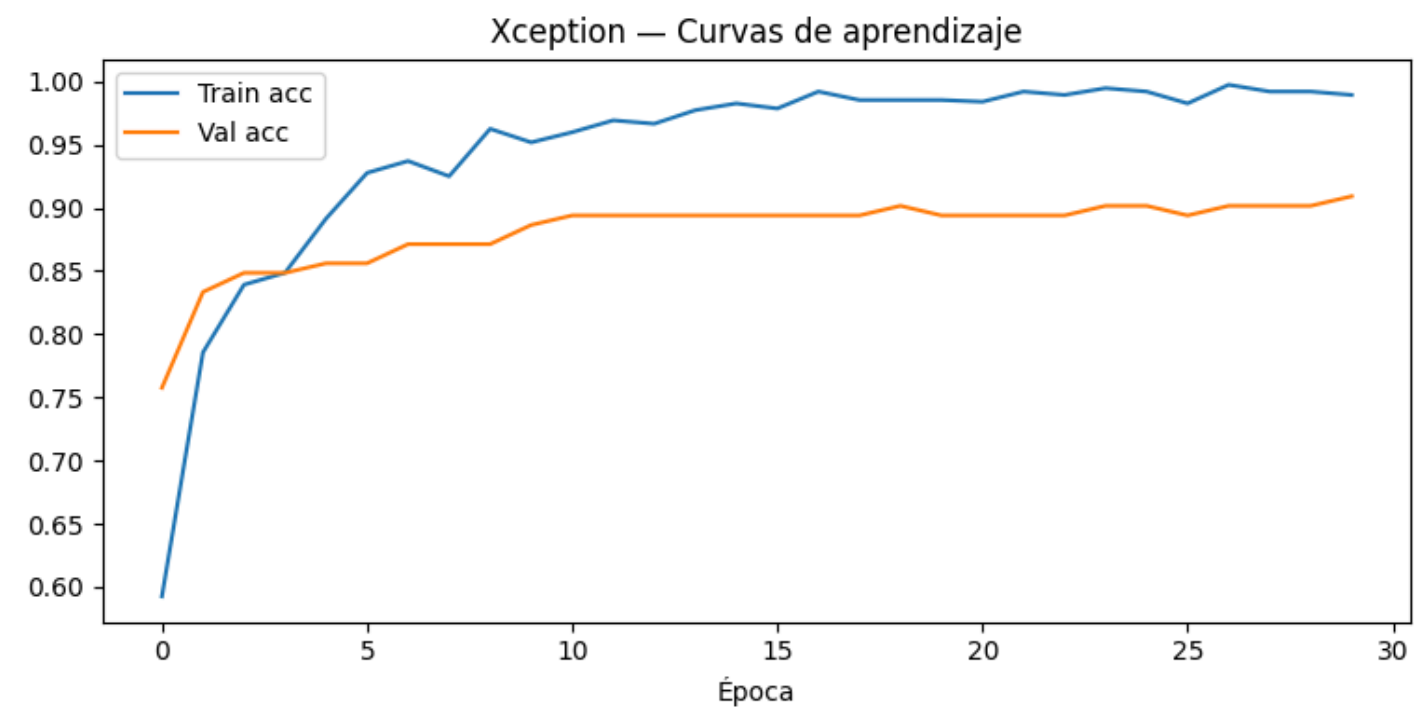
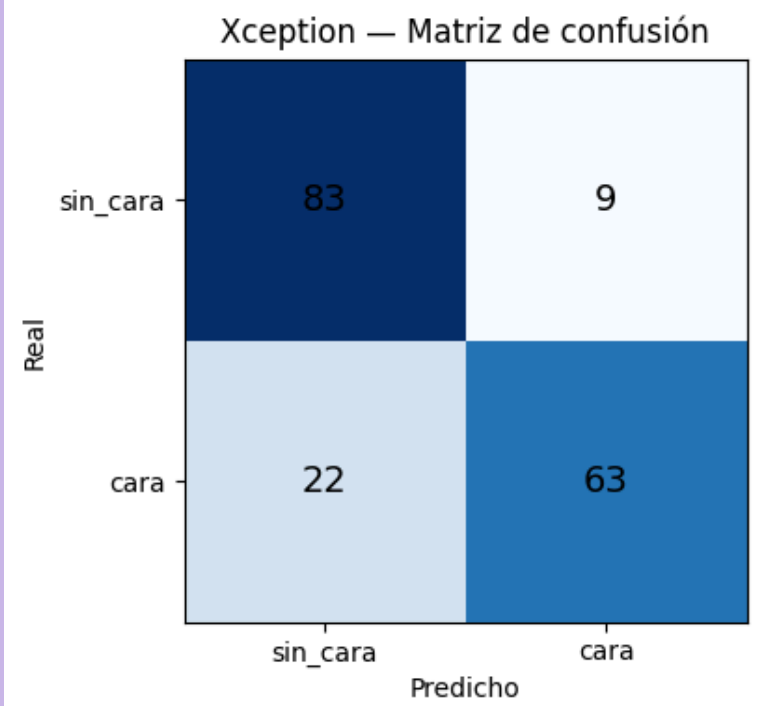
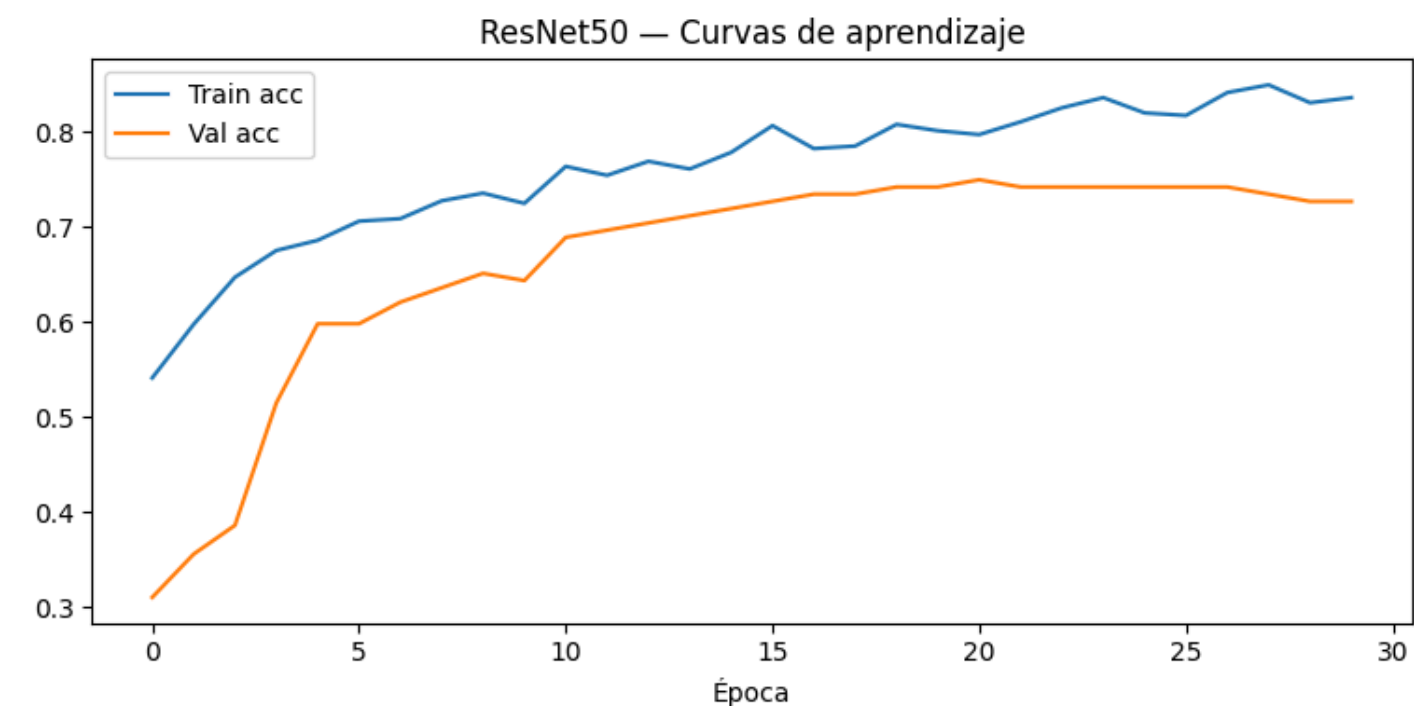
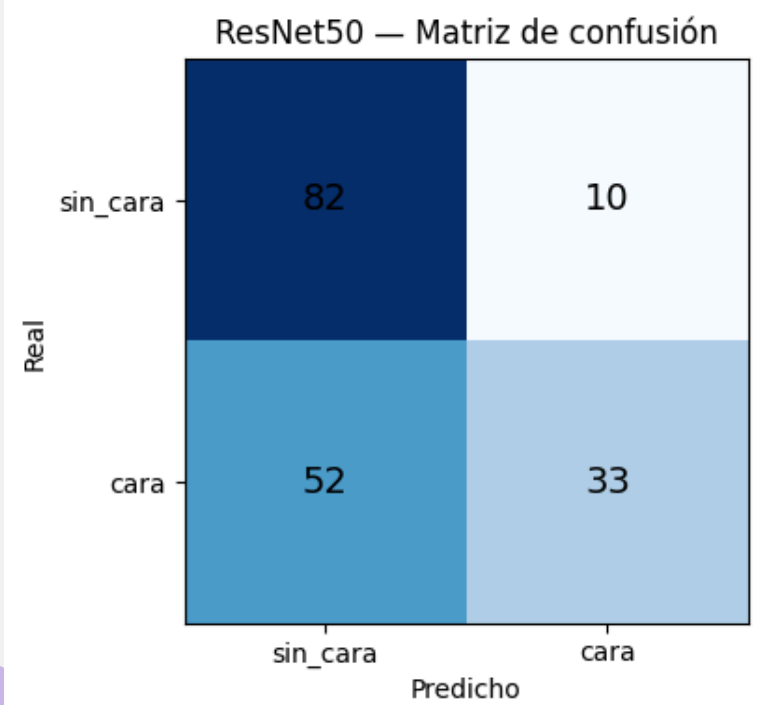
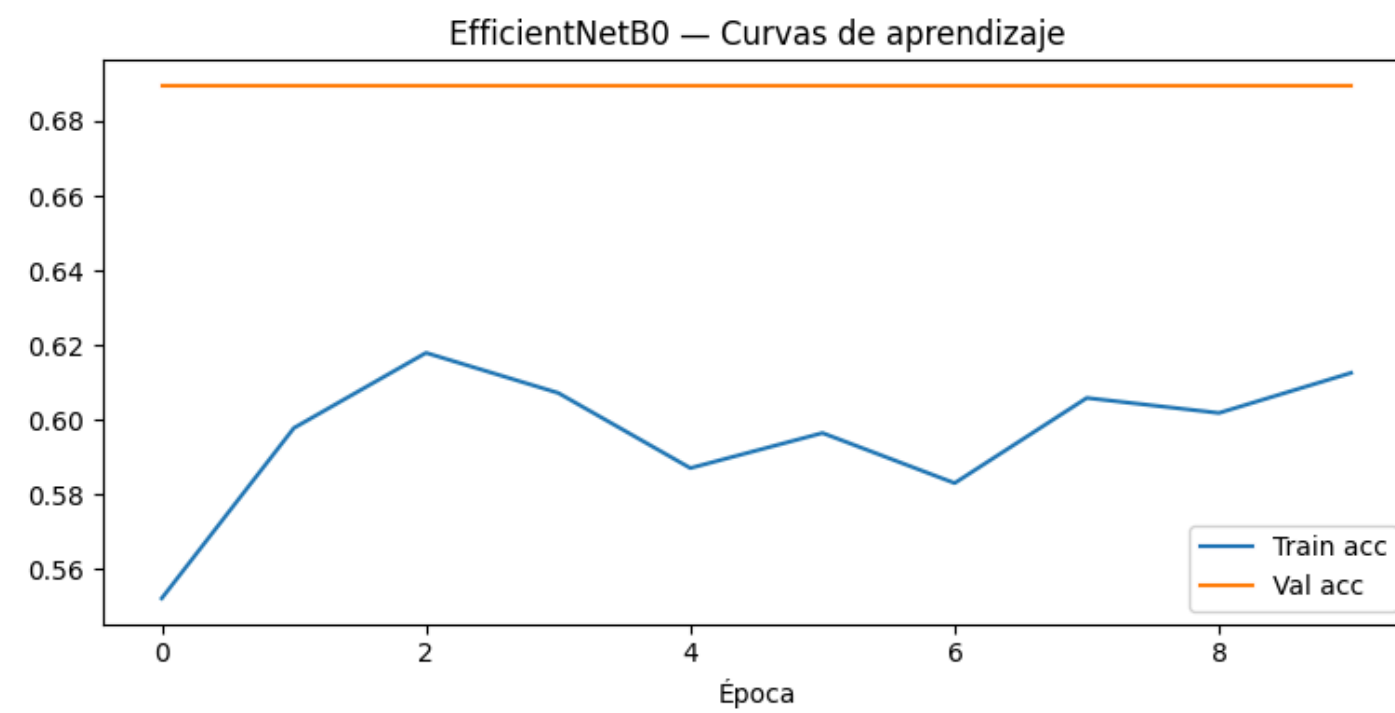
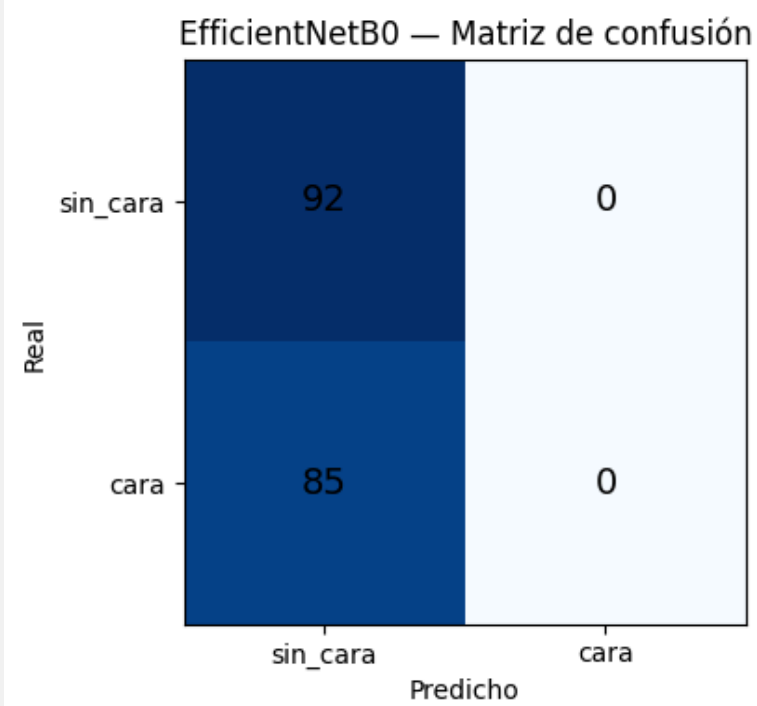
Mejoras de modelo



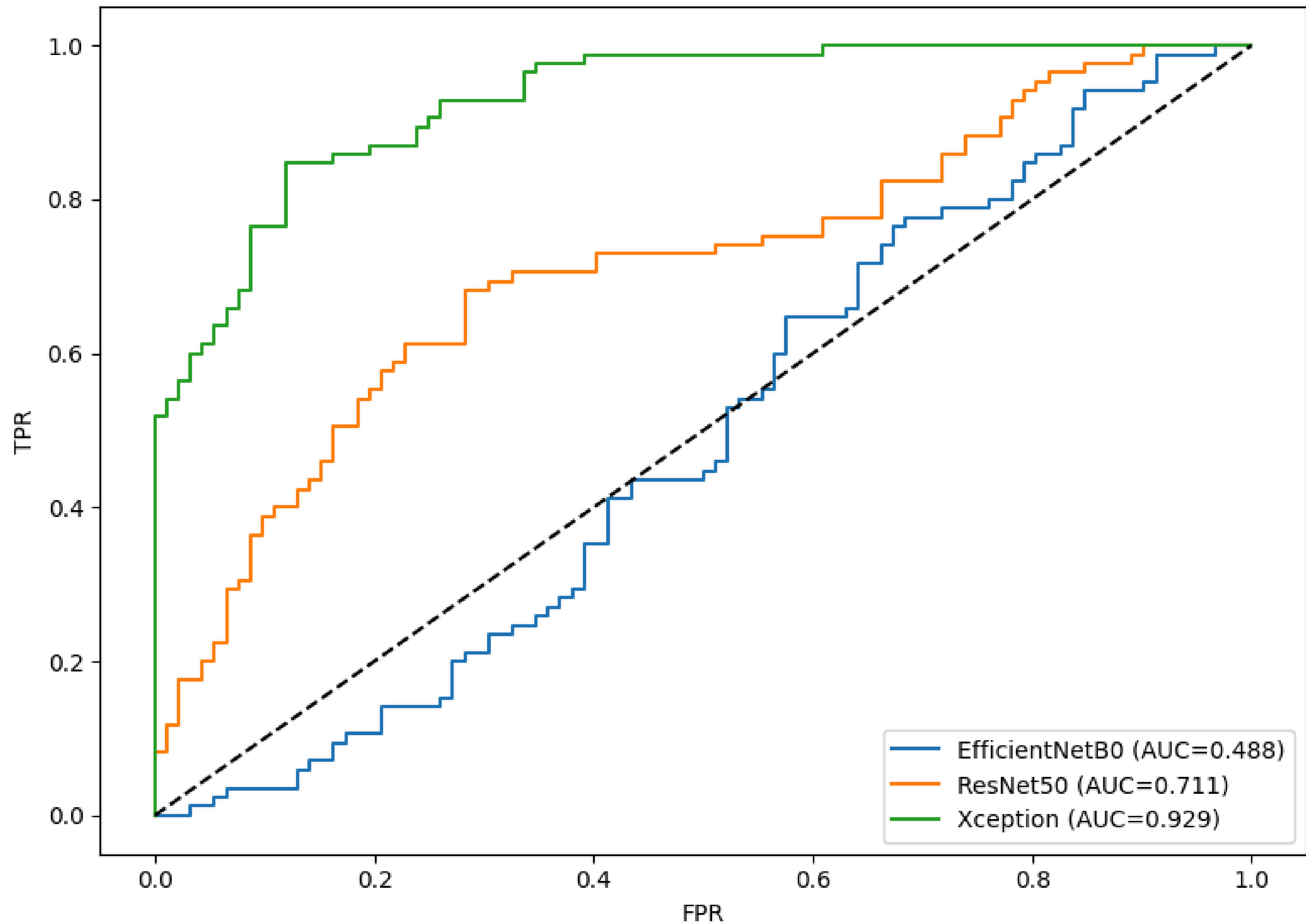
MODELOS PREENTRENADOS

Modelo	Input shape	Observaciones
EfficientNetB0	224×224	Ligero y eficiente, buen baseline
ResNet50	224×224	Robusto, estable en transfer learning
Xception	299×299	Basado en separable convolutions, muy bueno captando texturas y patrones locales





Curvas ROC — comparativa de modelos



MODELOS PREENTRENADOS

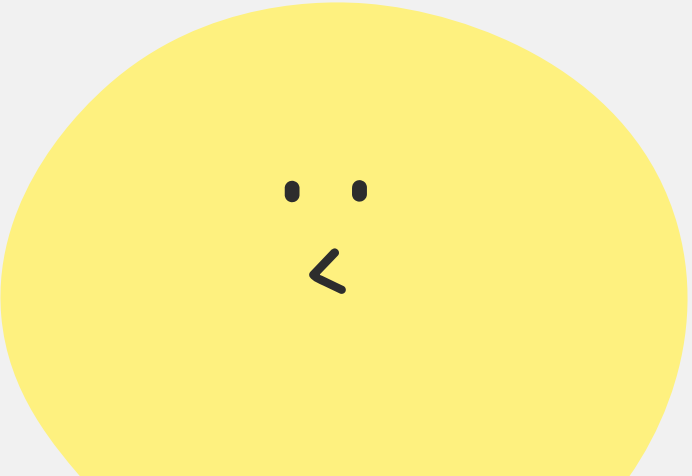
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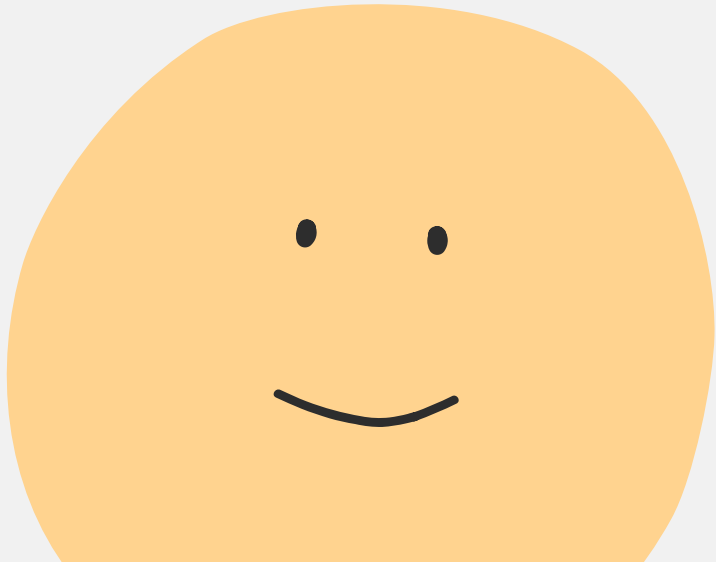


MEJORAS EN

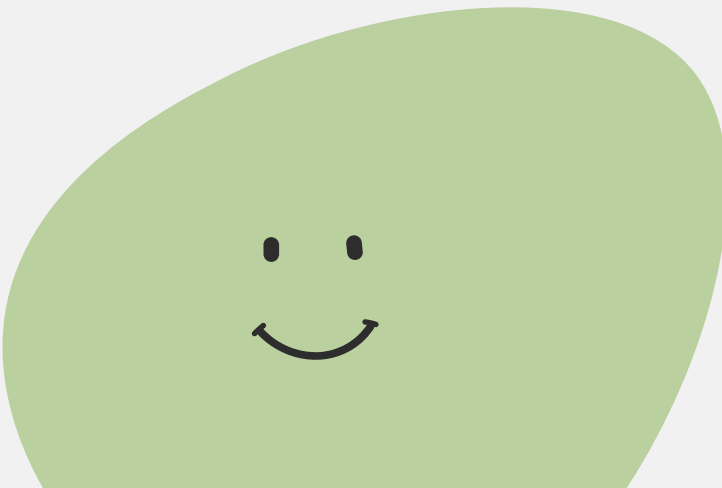
LOS MODELOS



<



☺



☺



☺

ESCALA DE GRISES

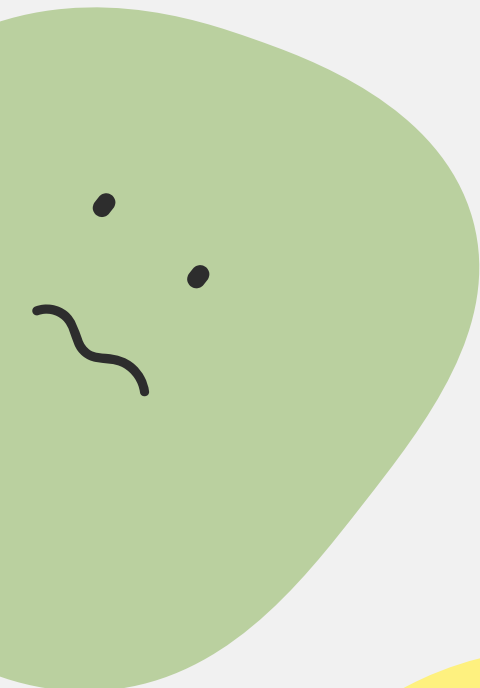
FINE-TUNING

AUGMENTATION

AUG + FT



CULO



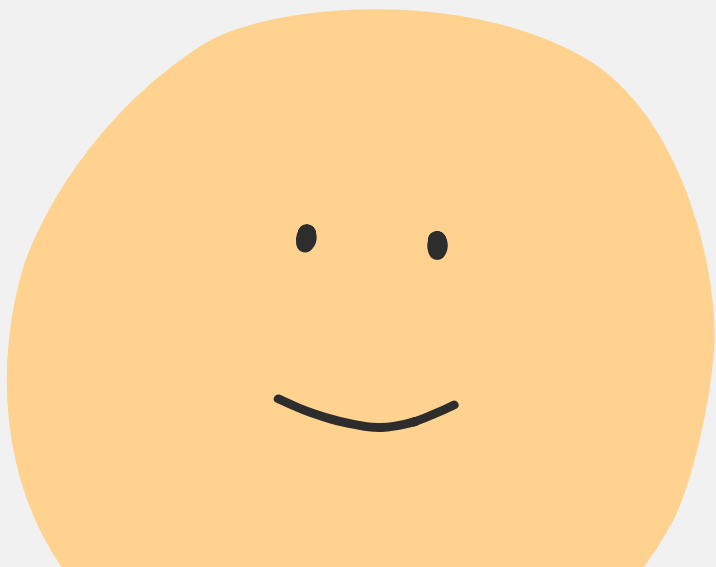
O



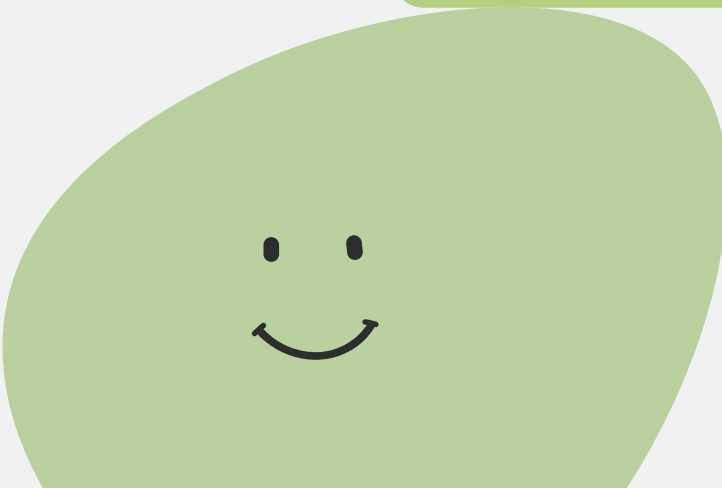
CODO



<



—



—



—



CARA ~~COO~~

O



NO CARA ~~COO~~

CARA — confianza: 99.99%



CARA — confianza: 98.16%



SIN CARA — confianza: 99.97%



CARA — confianza: 73.60%



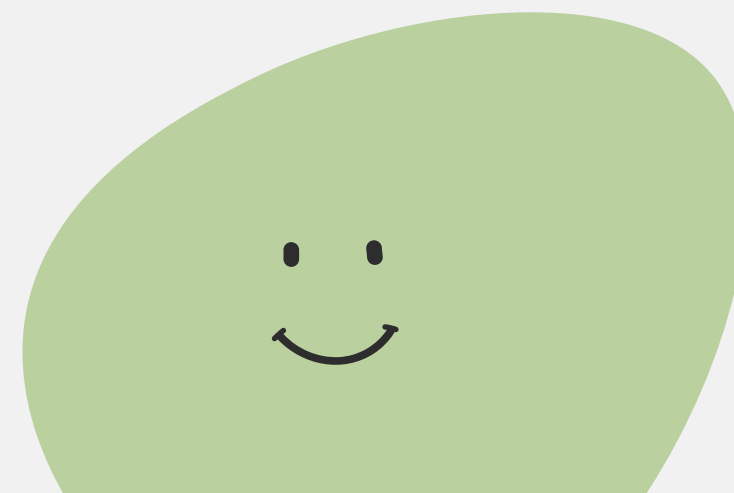
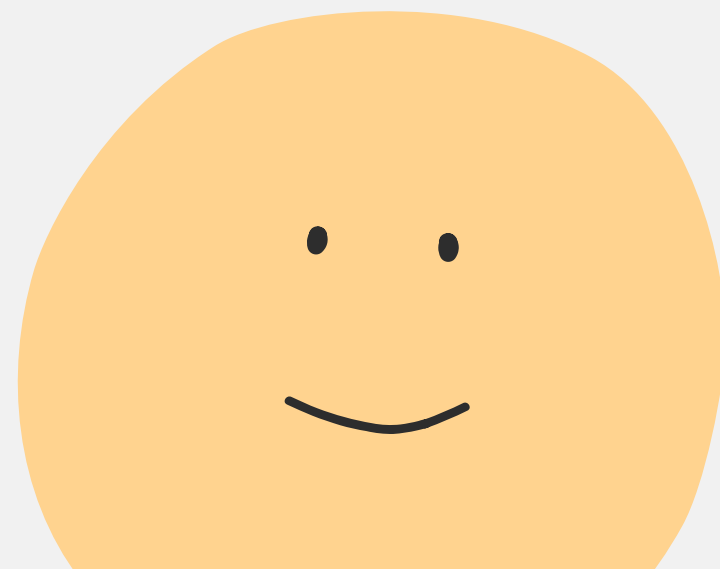
SIN CARA — confianza: 71.22%



CARA — confianza: 81.38%



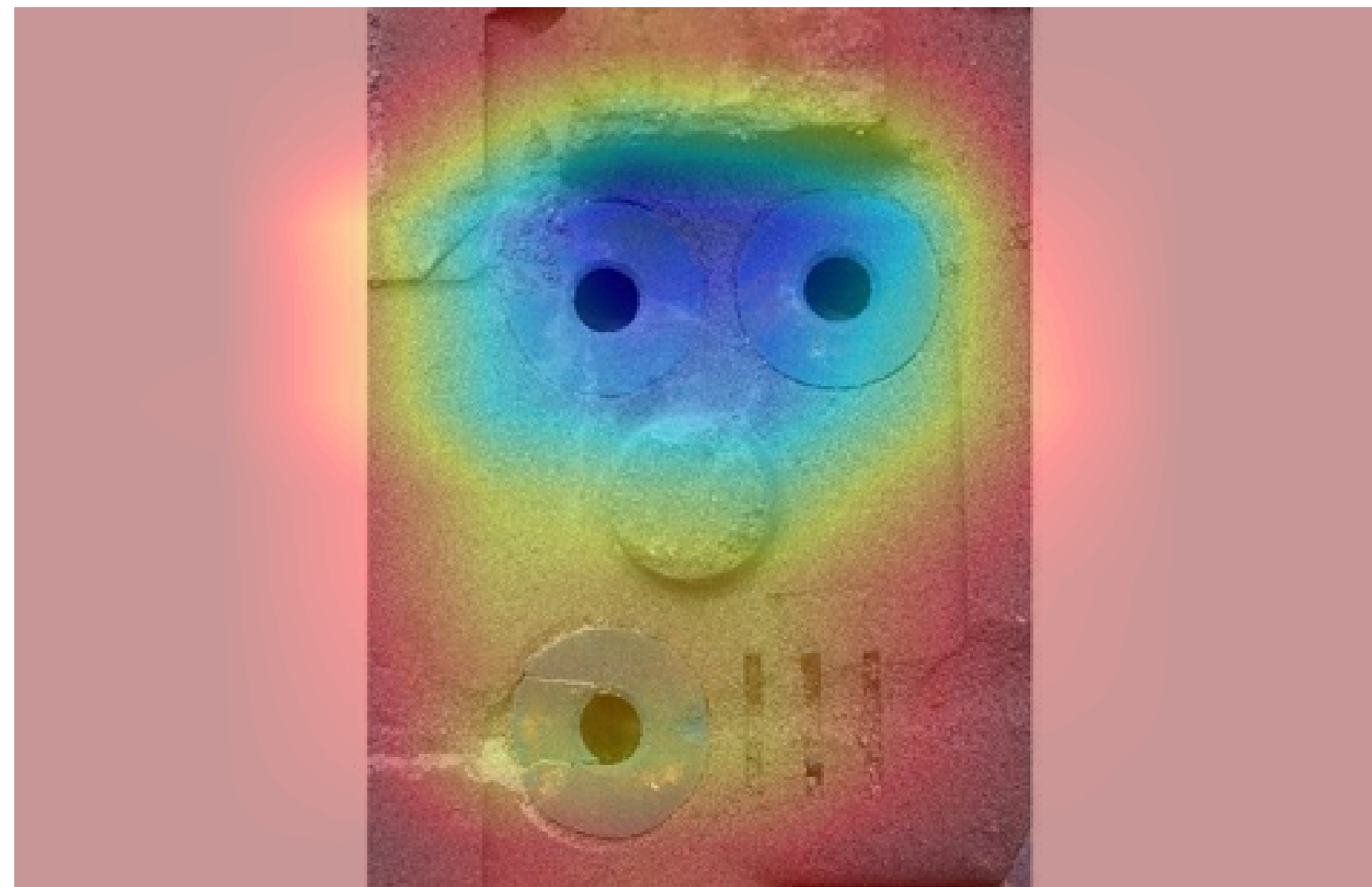
QUÉ ESTÁ
PASANDO??



CARA — confianza: 99.99%



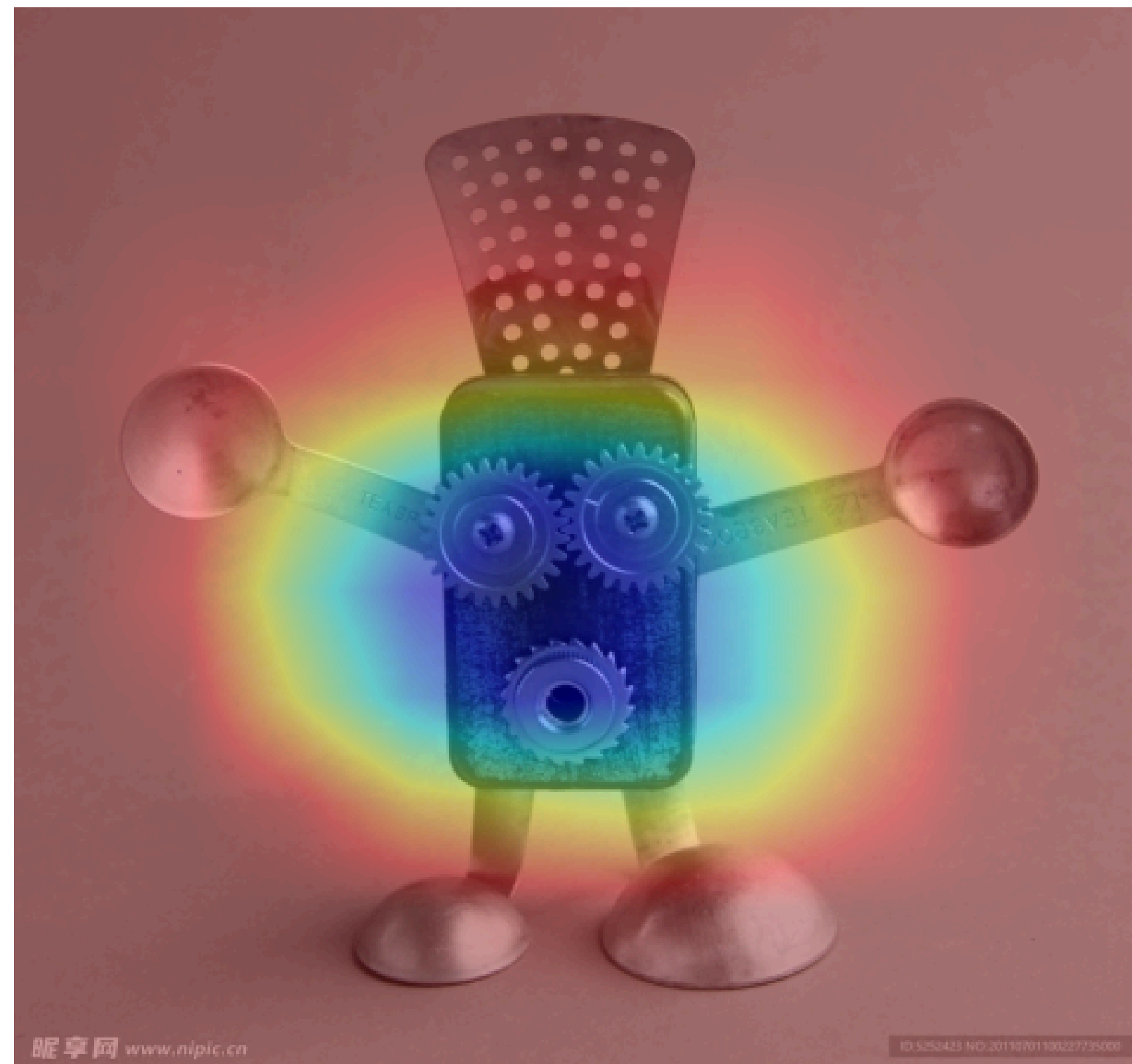
Grad-CAM



CARA — confianza: 98.16%



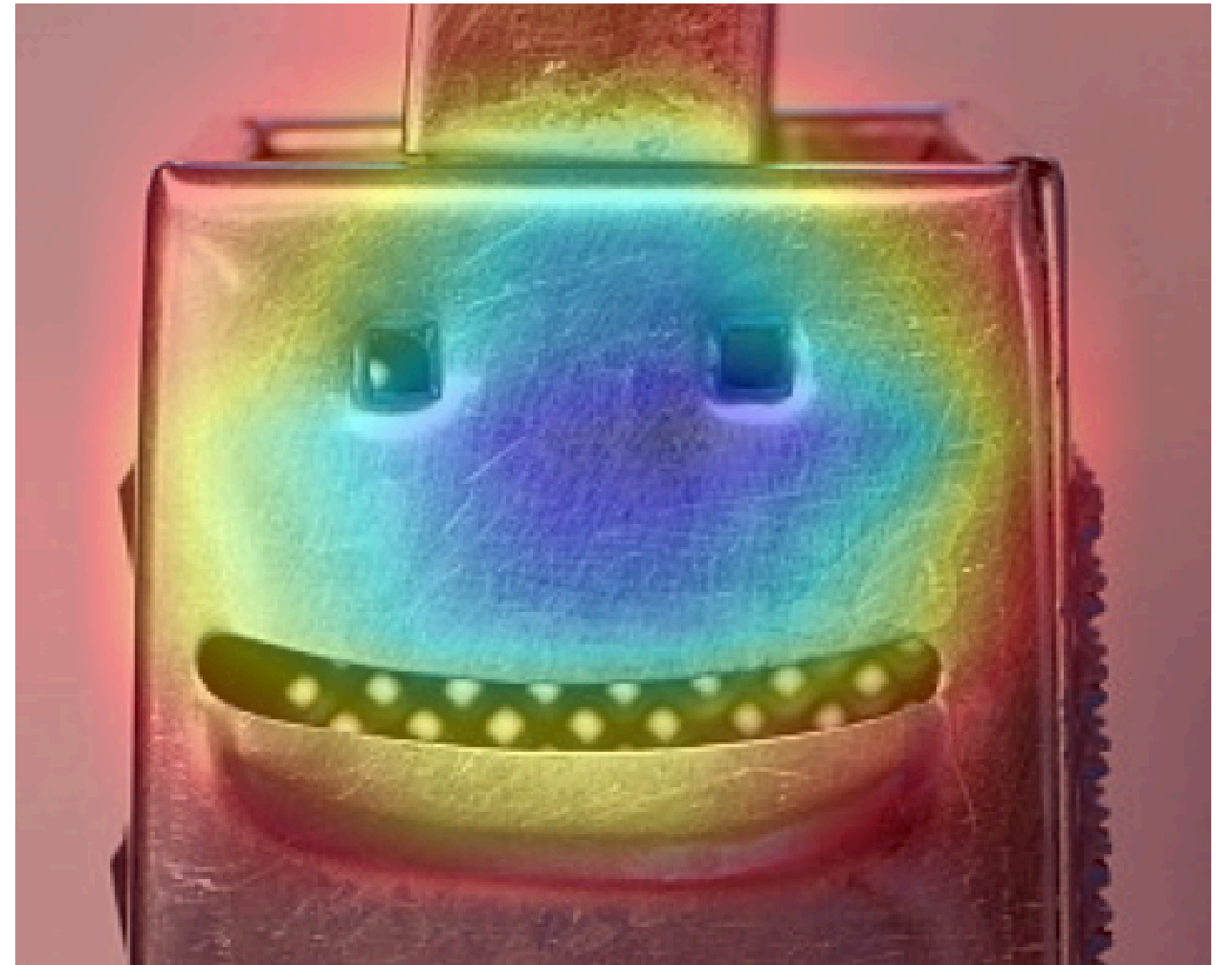
Grad-CAM



CARA — confianza: 100.00%



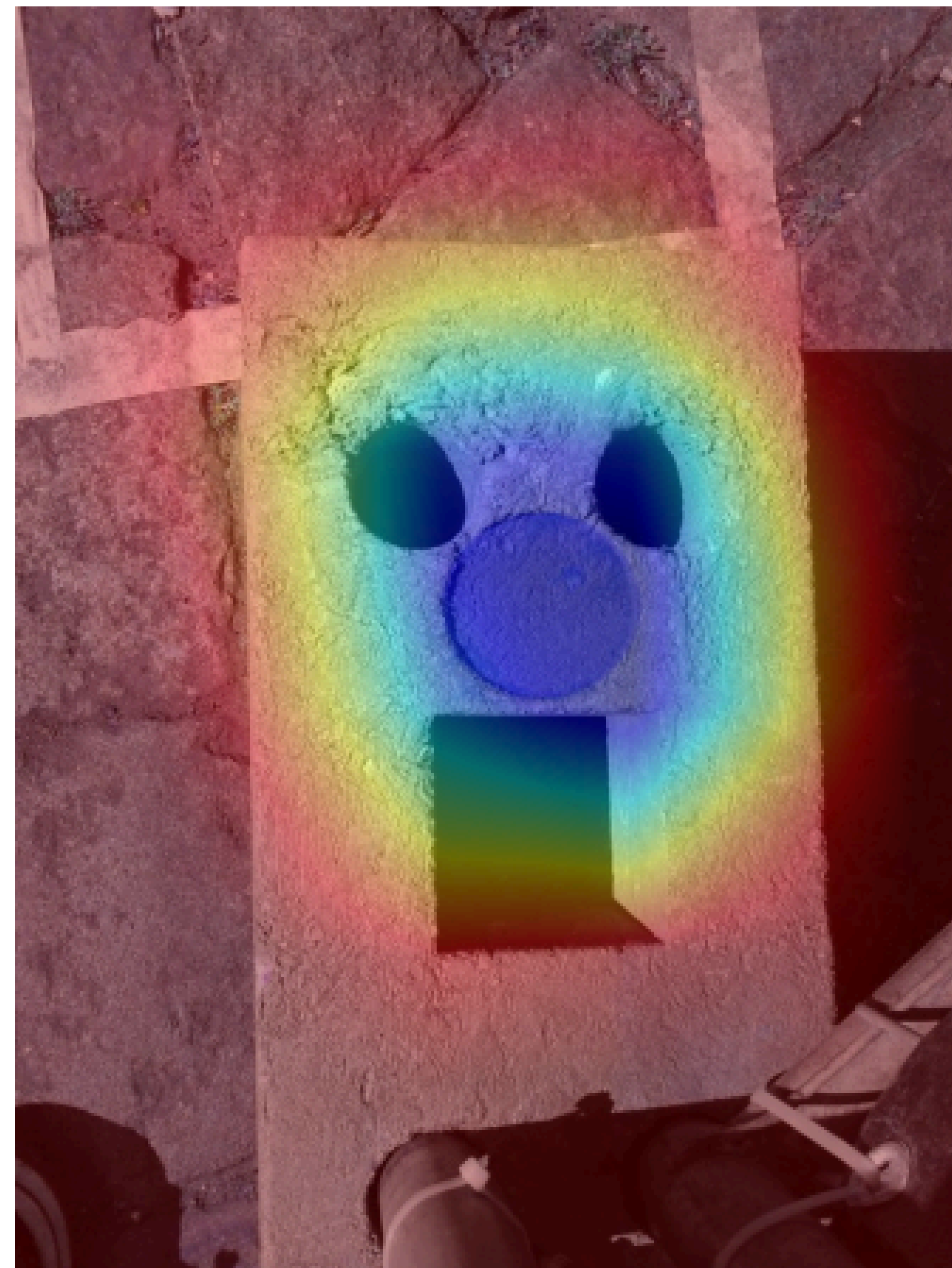
Grad-CAM



CARA — confianza: 99.57%



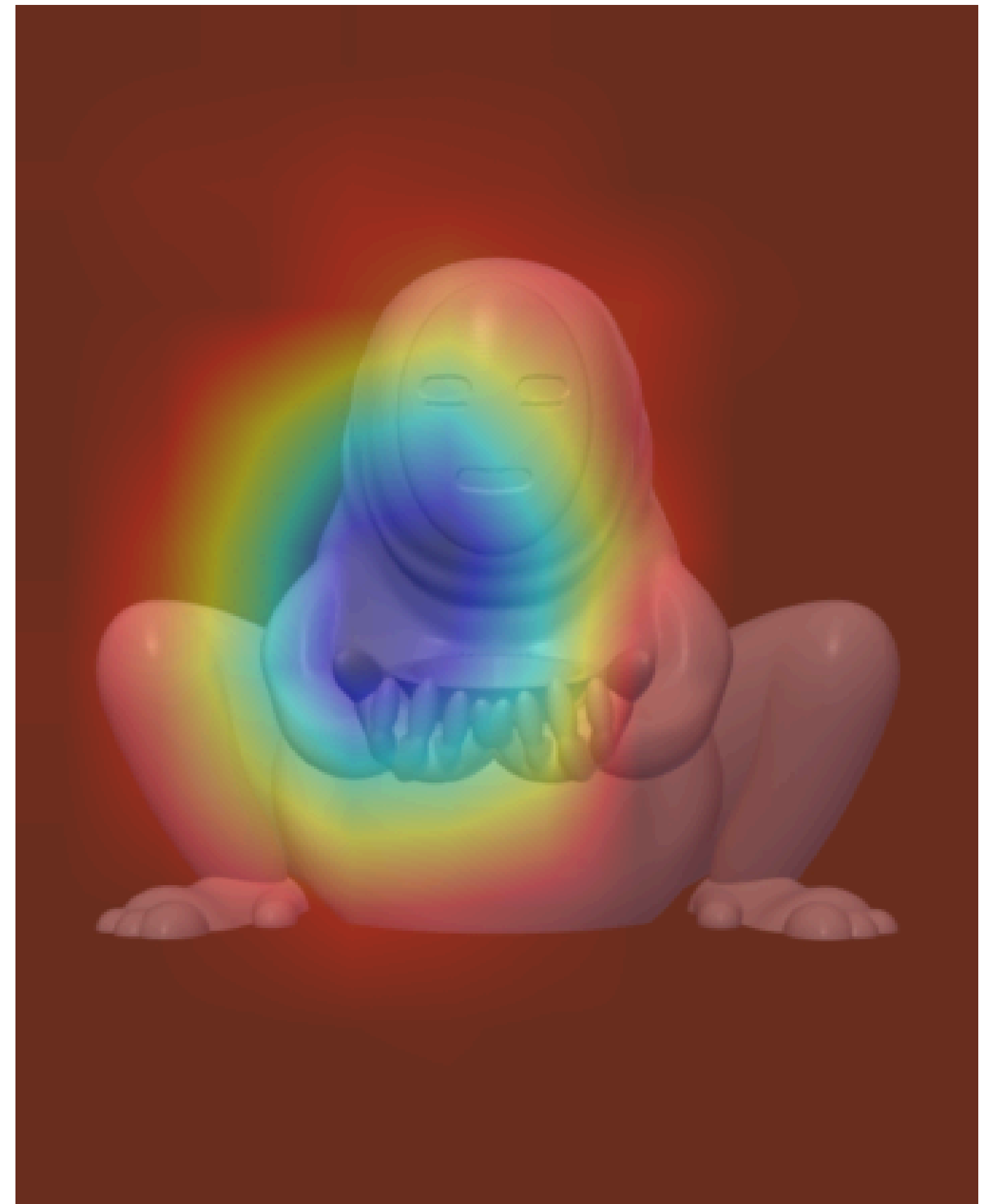
Grad-CAM



SIN CARA — confianza: 61.80%



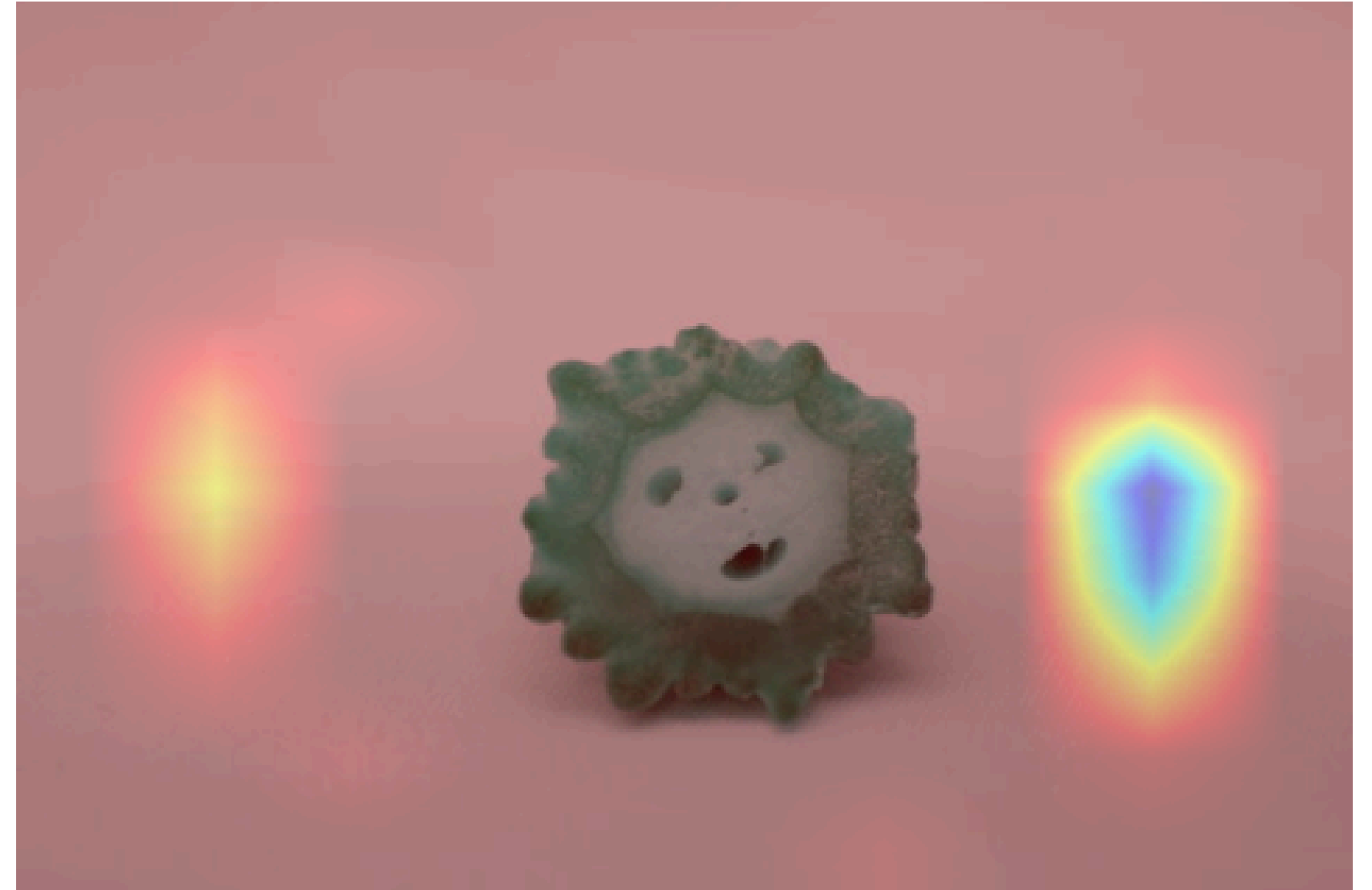
Grad-CAM



SIN CARA — confianza: 96.82%



Grad-CAM



SIN CARA — confianza: 71.22%



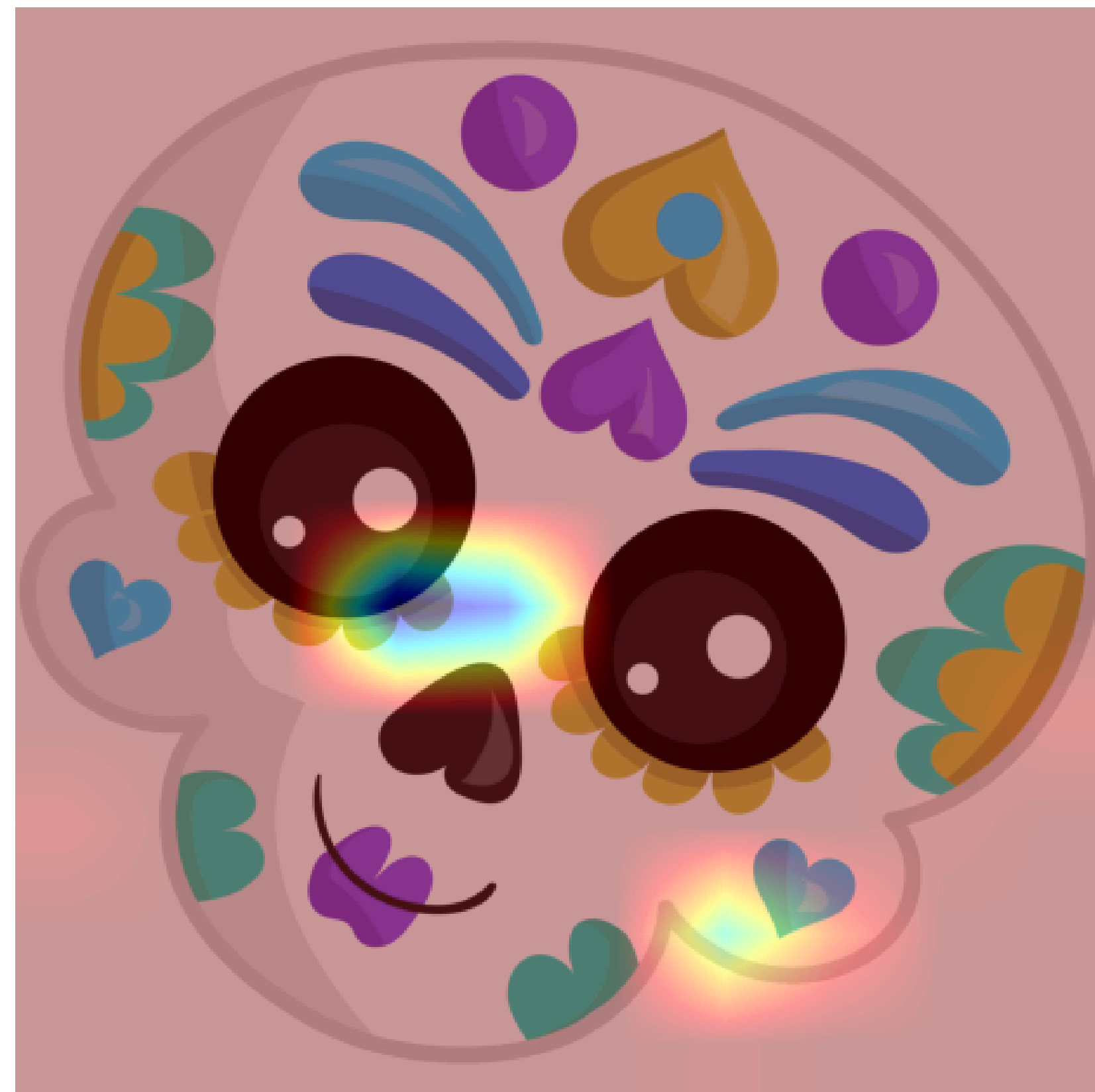
Grad-CAM



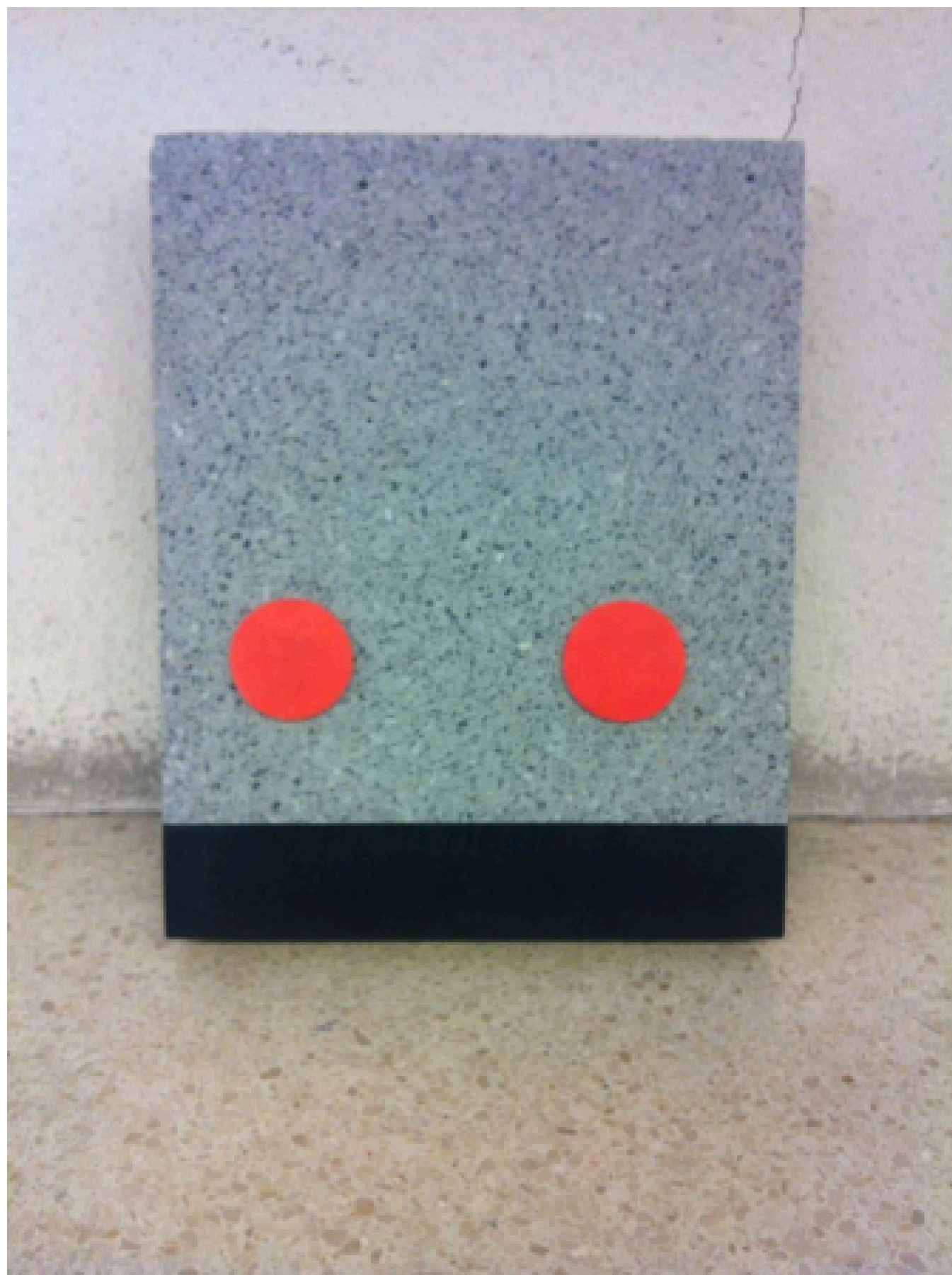
SIN CARA — confianza: 75.57%



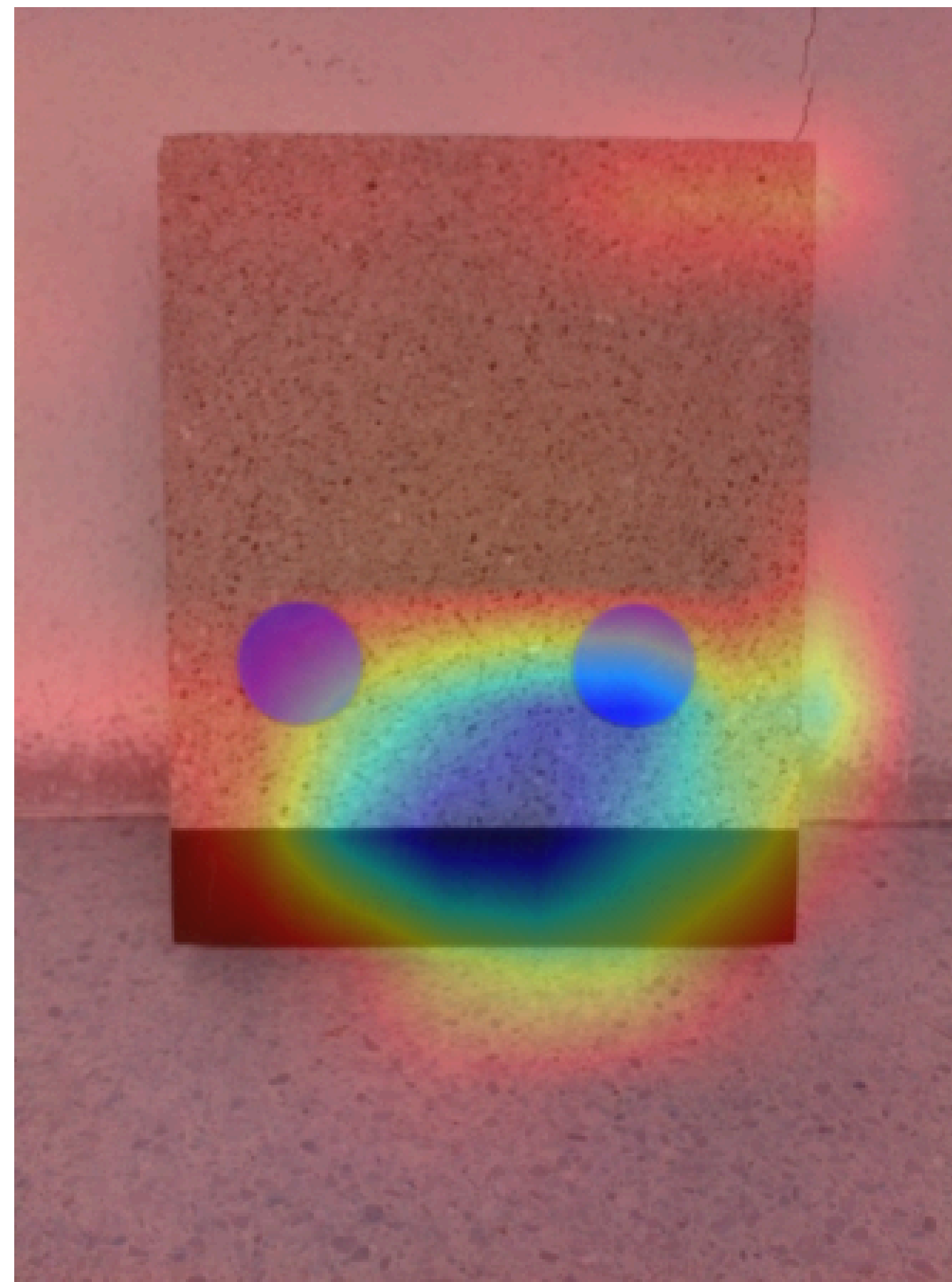
Grad-CAM



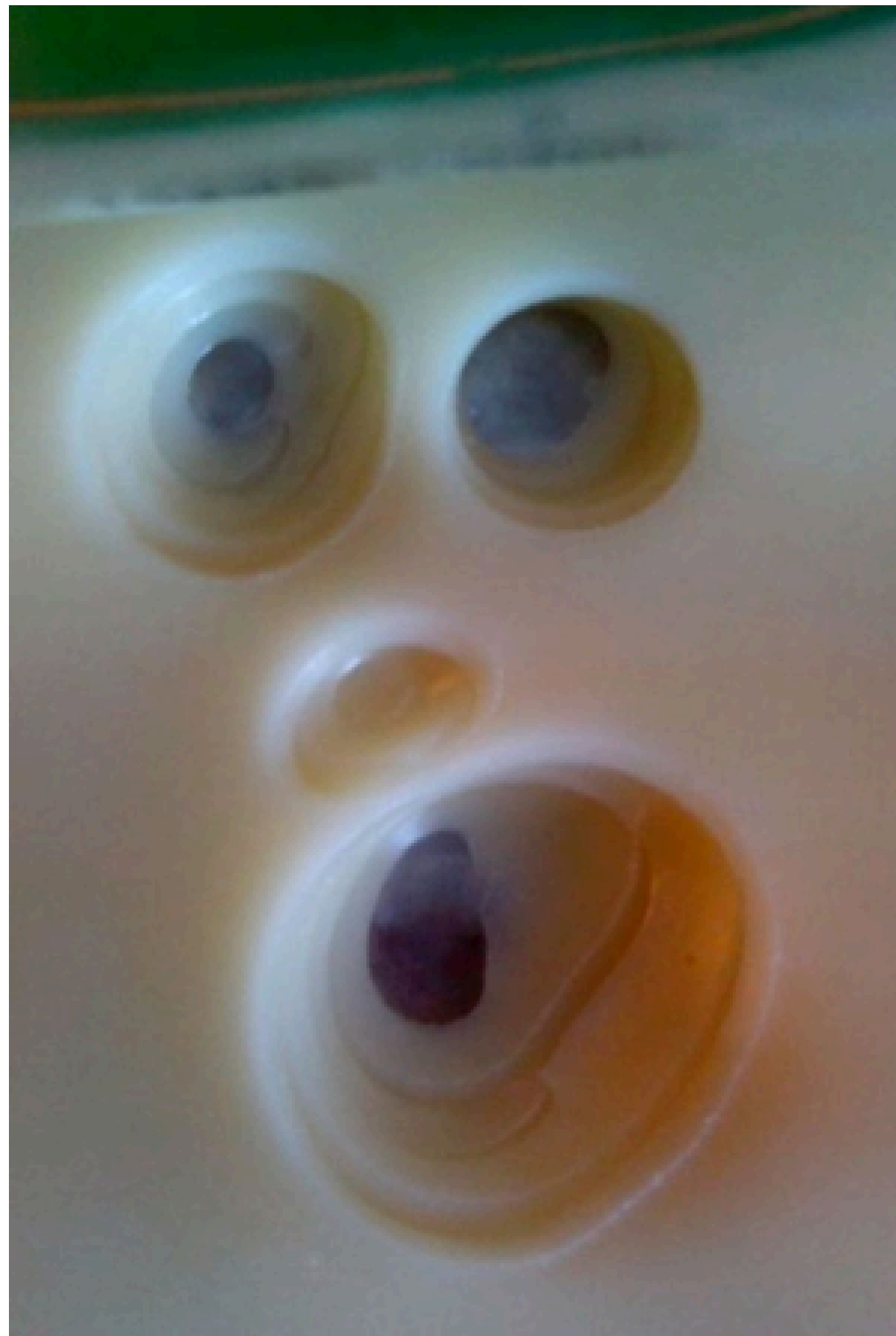
SIN CARA — confianza: 56.83%



Grad-CAM



CARA — confianza: 57.69%



Grad-CAM



SIN CARA — confianza: 65.72%



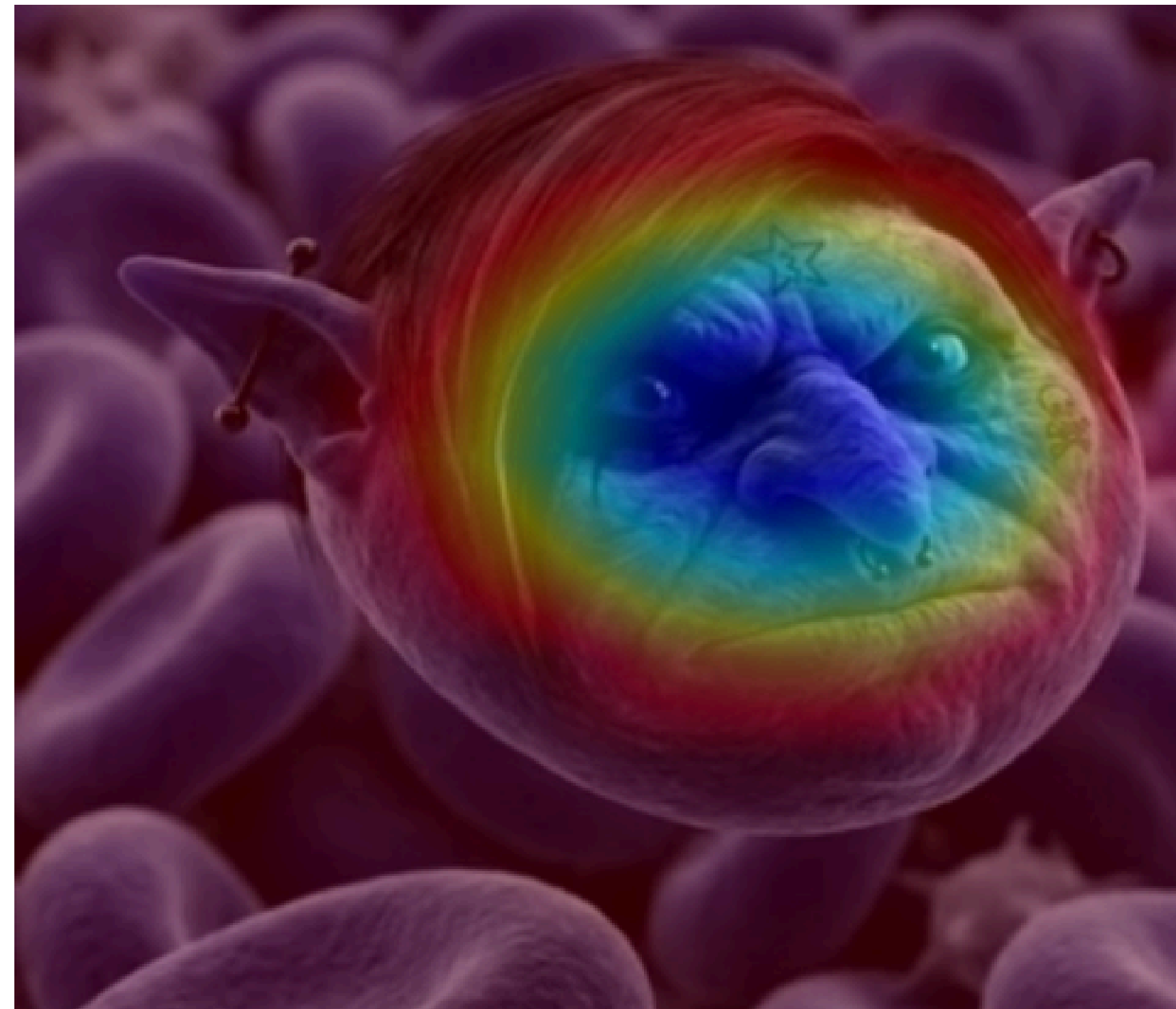
Grad-CAM



CARA — confianza: 97.31%



Grad-CAM



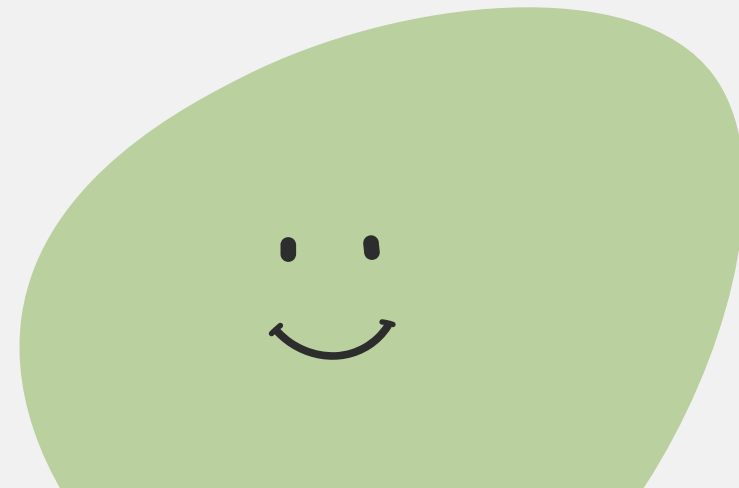
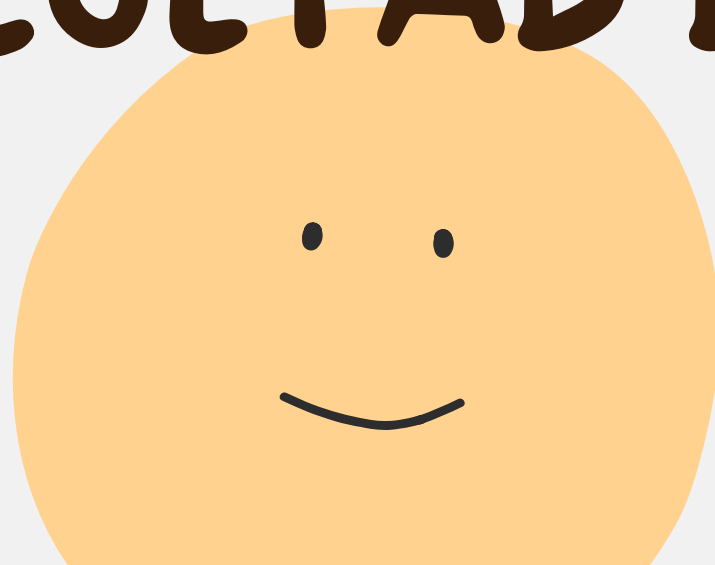
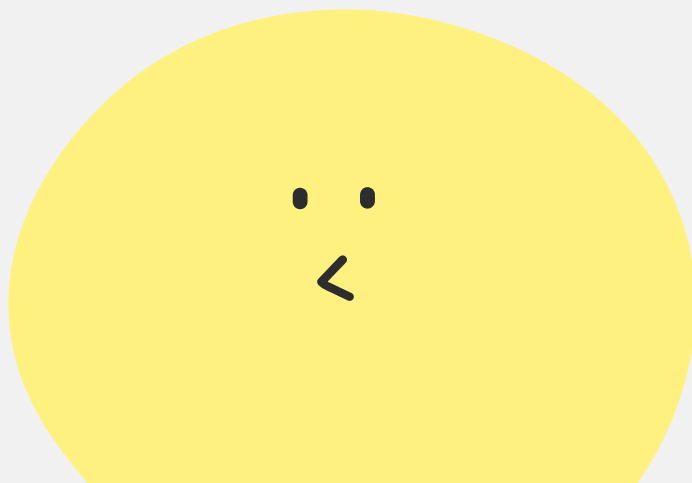
CONCLUSIONES

XCEPTION ES EL MEJOR MODELO

GRISES + AUGMENTATION + FT

GRAD-CAM

DIFICULTAD DE FORMAS DIVERSAS



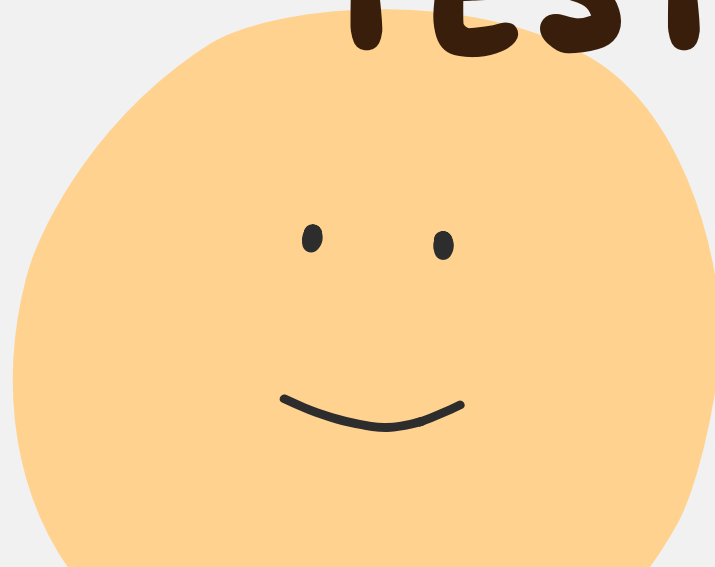
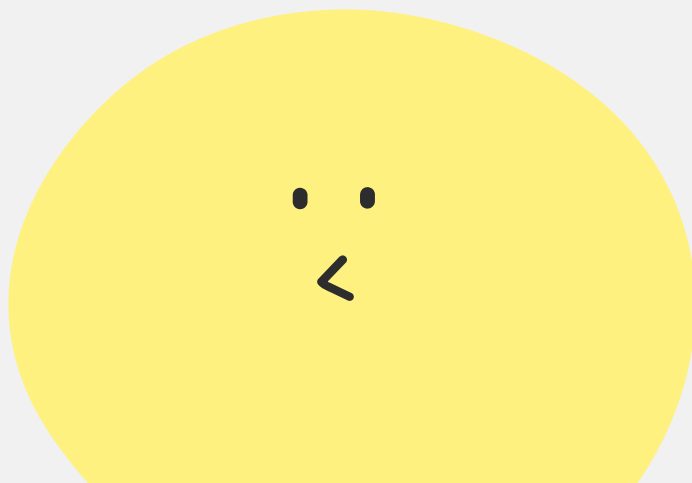
MEJORAS FUTURAS

ENSEMBLE DE MÚLTIPLES MODELOS

THRESHOLD AUTOMÁTICO

API

TESTING A/B





THANK YOU VERY
MUCH

